

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-1393
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how EduGenie can be extended to support a larger user base, higher request volumes, additional data, stronger security controls and new educational features.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Dependence on an external Gemini API for AI-generated responses.	Response availability and latency can depend on the external AI service.	High
2	Current implementation does not require a persistent application database.	Long-term learner progress and history are not stored as a core feature.	Medium
3	Higher concurrent usage can increase backend and AI-service load.	Response time may increase as concurrent requests grow.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single deployed web application with FastAPI	Introduce a managed database for user profiles, learning history, quiz results and progress.
2	Data Storage	Persistent database is not required by the current	Implement caching where appropriate, asynchronous/background processing and monitoring to
3	Performance	AI requests are processed through the backend and external Gemini service.	Secure secret management, authentication, authorization, rate limiting, HTTPS and
4	Security	API credentials are intended to be supplied through configuration.environment	Secure secret management, authentication, authorization, rate limiting, HTTPS and

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve UI, error handling, API monitoring and deployment reliability	0–3 months	Better usability and operational stability.
Phase 2	User accounts, database-backed learning history and progress tracking	3–6 months	Personalized learning history and progress
Phase 3	Multilingual support, voice interaction and richer image/PDF learning inputs	6–12 months	Broader accessibility and more learning for
Phase 4	Mobile application, dashboards, gamification and learning-platform integration	12+ months	Expanded learning ecosystem and engagement capabilities

Future Roadmap – Continued

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Scalability Strategy Summary

The roadmap prioritizes improving reliability and monitoring first, then adding persistent learning data and personalization, followed by broader input methods and mobile/learning-platform integration. These are proposed future enhancements rather than features claimed as currently implemented.